

DESIGN & ANALYSIS OF ALGORITHMS

BCSE204L | Previous Year Questions Bank

7

Papers Analyzed

70+

Questions Collected

10

Modules Covered

2022–26

Years Covered

Paper	Key Topics
Undated Paper	Recursion, Knapsack, Divide & Conquer, KMP, Floyd-Warshall, Ford-Fulkerson, TSP, NP
Winter 2022-23	Karatsuba, Knapsack, Huffman, Matrix Chain, KMP/LPS, Ford-Fulkerson, Graham Scan
April 2023 (G2+TG2)	Recurrence, Fractional Knapsack, Matrix Chain, Rabin-Karp, Bellman-Ford, Convex Hull
May 2024 (A1+TA1)	Matrix Chain, KMP, Floyd-Warshall, Graham Scan, Quicksort, NP, Edmonds-Karp, N-Queens
May 2024 (A2+TA2)	Recurrence, Edit Distance, KMP, Bellman-Ford, Jarvis March, Quicksort, Ford-Fulkerson
April 2025 (A1+TA1)	Karatsuba, Huffman, Held-Karp TSP, Branch & Bound, KMP+Rabin-Karp, Bellman-Ford
April 2026 (A1+TA1)	Fractional Knapsack, Master Theorem, Max Subarray, N-Queens, TSP, Rabin-Karp, Floyd-Warshall

MOST REPEATED TOPICS (appear in 5+ papers)

Ford-Fulkerson / Max Flow	7/7
Knapsack (0-1 & Fractional)	7/7
Floyd-Warshall	6/7
KMP String Matching	6/7
Matrix Chain Multiplication	5/7

Convex Hull (Graham/Jarvis)	5/7
Randomized Quicksort	5/7
NP-Completeness / 3-CNF SAT	5/7

Module 1: Algorithmic Analysis & Recurrence Relations

CO1 | 6 Questions

Asymptotic Notation (Big-O, Omega, Theta) | Recurrence Relations | Recursive Tree Method | Master Theorem

REPEATED TOPIC

Q1

10M

Obtain an upper bound for $T(n) = 3T(n/4) + n^2$ using the recursive tree method and verify using Master's theorem.

[Undated] [Apr 2026]

Tip: This EXACT question appeared in 2 papers. Master's theorem case 3 applies here.

Q2

4M

Solve $T(n) = 3T(n/3) + O(n^{1/2})$ and $T(n) = 4T(n/2) + n^2$ using the Master method.

[Apr 2023]

Q3

4M

Solve the recurrence $T(n) = 4T(\sqrt{n}) + \log^2 n$ using Master method.

[May 2024 B]

Q4

6M

Write a non-recursive algorithm to find the sum of first n odd numbers. Using loop invariant property, prove the correctness of the algorithm.

[Apr 2023]

Q5

10M

Design an algorithm for sum of all elements of a 2D matrix and analyze its time complexity. How to prove correctness using Induction method.

[May 2024 A]

Q6

6M

Write an iterative algorithm to find sum of $1/i(i-1)$ for $i=2$ to n . Prove your algorithm is correct.

[May 2024 B]

Module 2: Divide and Conquer

CO1 | 5 Questions

Karatsuba Multiplication | Maximum Subarray (D&C;) | Merge Sort | Quick Sort

REPEATED TOPIC

Q1

10M

How does Karatsuba algorithm handle numbers with odd number of digits? Write the algorithm/program and show stepwise execution for binary numbers 10101 and 11. Analyse time complexity.

[May 2024 A]

Tip: Karatsuba appeared in 2 papers. Know the 3-multiplication trick.

REPEATED TOPIC

Q2

10M

The Karatsuba algorithm is a divide-and-conquer approach for fast integer multiplication. Break it into stages: problem decomposition, recursive subproblems, and recombination. Derive and analyse its time complexity comparing with standard multiplication. Illustrate for 21×143 .

[Apr 2025]

REPEATED TOPIC

Q3

10M

Design a Divide and Conquer algorithm to find the Maximum Subarray whose sum is maximum. Consider array $[2, -4, 1, 9, -6, 7, -3]$. Trace the steps with recursive tree. Discuss time complexity.

[Undated]

Tip: Classic D&C; problem. Know how to split into left/right/cross subarray.

REPEATED TOPIC

Q4

10M

Design the pseudocode for Divide and Conquer Maximum Subarray. Apply algorithm on $A = [-2, 5, -1, 3, -4, 7, -2, 1]$. Show the left sum, right sum, and cross sum for each recursive division. State the starting and ending indices of the maximum subarray and its sum.

[Apr 2026]

Q5

10M

You are given two DNA sequences DNA1 and DNA2. Find the minimum number of single-character edits (insertions, deletions, or substitutions) to transform DNA1 into DNA2. Given DNA1='AGTACG' and DNA2='GTCAGT', minimum edits = 4. Develop a DP algorithm for minimum edit distance.

[May 2024 B]

Module 3: Greedy Algorithms

CO2 | 9 Questions

Fractional Knapsack | Activity Selection | Huffman Coding | Job Scheduling

REPEATED TOPIC

Q1

10M

Solve the Fractional Knapsack problem using Greedy strategy. Knapsack capacity is 35. Weights = [10, 20, 12], profits = [60, 100, 120]. Find optimal items to include.

[Undated]

Tip: Sort by profit/weight ratio. This exact instance appeared multiple times.

REPEATED TOPIC

Q2

10M

State the fractional Knapsack problem. Use greedy method for $n=7$, capacity $W=23$. Profits = [29, 17, 20, 19, 6, 28, 42]. Weights = [7, 5, 6, 3, 9, 2, 5]. Show all steps.

[Apr 2023]

REPEATED TOPIC

Q3

5M

Design an algorithm for the Fractional Knapsack problem using Greedy approach. Profits=[5,10,15,7,8,9], Weights=[1,3,5,4,1,3], Capacity $W=15$. Show profit/weight ratio calculations and selection order. Why is Fractional Knapsack solved using Greedy while 0/1 Knapsack cannot be?

[Apr 2026]

REPEATED TOPIC

Q4

5M

Validate the statement: 'Greedy approach does not ensure an optimal solution for 0-1 Knapsack problem' using two examples — knapsack capacity 8 (weights [3,4,6,5], values [20,30,10,40]) and capacity 20 (weights [15,10,10], values [9,5,5]).

[Winter 2022-23]

REPEATED TOPIC

Q5

10M

Construct the optimal Huffman code for characters a-h with frequencies derived from first 8 prime numbers [2, 3, 5, 7, 11, 13, 17, 19]. Derive time and space complexity. Show Huffman coding takes less space than fixed size code.

[Apr 2025]

Tip: Huffman appeared in 3 papers. Build tree bottom-up, always merge two smallest.

Q6

5M

Two encoding schemes E1 and E2 encode characters R,G,B,Y as (0,011,111,11) and (0,11,101,100) respectively. String 'RGRBYRR' is encoded using both. Which is Huffman encoding? Justify.

[Winter 2022-23]

REPEATED TOPIC

Q7

10M

You are attending a workshop with various sessions. Find a greedy approach to attend maximum sessions without conflicts. Sessions: A(9:00-10:00), B(9:30-11:30), C(11:00-12:00), D(12:30-14:00), E(13:00-14:30), F(14:30-15:00).

[May 2024 B] [Apr 2026]

Tip: Activity Selection — sort by end time, pick non-overlapping greedily.

Q8

10M

Define Greedy Knapsack. $n=7$, $M=15$, profits=[10,5,15,7,6,18,3], weights=[2,3,5,7,1,4,1]. Find optimal solution.

[May 2024 A]

Q9

10M

Solve tasks with deadlines and profits to produce maximum profit. Jobs J1-J5, Deadlines=[2,2,1,3,4], Profits=[20,60,40,100,80].

[May 2024 A] [May 2024 B]

Module 4: Dynamic Programming

CO2 | 7 Questions

Matrix Chain Multiplication | Travelling Salesman Problem (Held-Karp) | 0-1 Knapsack DP | Longest Common Subsequence

REPEATED TOPIC

Q1

10M

Design a dynamic programming algorithm for parenthesization of matrix chain which will require minimum number of scalar multiplications. Consider chains $A(2 \times 3)$ $B(3 \times 2)$ $C(2 \times 4)$ $D(4 \times 3)$. Illustrate the algorithm.

[Undated]

Tip: Matrix Chain is one of the most classic DP problems. Always in exams.

REPEATED TOPIC

Q2

10M

Outline important factors of Dynamic Programming and perform Matrix Chain Multiplication for $\{4, 10, 3, 12, 20, 7\}$. Matrices sizes: 4×10 , 10×3 , 3×12 , 12×20 , 20×7 . Compute $M[i, j]$, $0 \leq i, j \leq 5$.

[May 2024 A]

REPEATED TOPIC

Q3

10M

Given a sequence of matrices $\langle 8, 5, 10, 20, 5 \rangle$. Using Dynamic Programming find the minimum number of scalar multiplications needed and also write the optimal multiplication order.

[Apr 2023]

Q4

10M

Given DP pseudocode for matrix chain multiplication — explain the reasons for assigning specific initial and final values for i loop, k loop, and j variable. Fill the M table and find total scalar multiplications for 5 matrices $[4, 10, 3, 12, 20, 7]$.

[Winter 2022-23]

REPEATED TOPIC

Q5

10M

A logistics company wants to optimize the delivery route for a salesperson visiting 4 cities exactly once. Using Held-Karp Algorithm (Dynamic Programming), compute the minimum travel cost for given distance matrix and show step-by-step calculations. Compare DP vs brute-force complexity.

[Apr 2025]

Tip: Held-Karp TSP appeared in 3 papers. Know $g(S, i) = \min \text{ over } k \text{ in } S \text{ of } [g(S - \{i\}, k) + \text{cost}(k, i)]$.

REPEATED TOPIC

Q6

10M

Solve the Travelling Salesman Problem for the given graph using Dynamic Programming. Cost Matrix (cities A,B,C,D): A-B=20, A-C=42, A-D=35, B-C=30, B-D=34, C-D=12. Starting city: A. Show all calculations, determine optimal tour and its cost.

[Apr 2026]

REPEATED TOPIC

Q7

10M

Design a polynomial time approximation algorithm for the Travelling Salesman Problem. Illustrate the algorithm for the given graph (starting vertex 1). Prove it is a 2-approximation algorithm.

[Undated]

Module 5: Backtracking

CO2 | 4 Questions

N-Queens Problem | Subset Sum Problem | 0-1 Knapsack via Backtracking

REPEATED TOPIC

Q1

10M

Explain N-Queens algorithm and explain with 4x4 chessboard, numbering rows and columns 1 through 4, in a State Space Tree using Backtracking (step-by-step). Also derive Time Complexity of N Queens using backtracking.

[May 2024 A]

Tip: N-Queens is a standard backtracking problem. Draw the state space tree.

REPEATED TOPIC

Q2

10M

Consider the N-Queens problem for $N=6$. Construct the state-space tree up to level 3. Clearly indicate promising and non-promising nodes. Show the complete backtracking process to obtain one valid solution. Analyse worst-case time complexity.

[Apr 2026]

REPEATED TOPIC

Q3

10M

Write the code snippet / pseudocode for implementing backtracking technique to find the subset of elements from a given set, whose sum adds up to a given target value. Find the first two subsets from the set [10,20,50,60,80] that sum up to target 80.

[Winter 2022-23]

Tip: Subset sum backtracking — know when to prune (current sum > target).

REPEATED TOPIC

Q4

10M

Define the Subset Sum problem. Solve it for the set {40,30,50,10,20} with a target sum of 60. Use backtracking approach.

[May 2024 B]

Module 6: Branch and Bound

CO2 | 3 Questions

0-1 Knapsack using FIFO Branch & Bound | Job Scheduling via Branch & Bound | State Space Tree with Cost/Bound Values

REPEATED TOPIC

Q1

10M

Solve the following instance of Knapsack problem using fixed length FIFO branch and bound strategy where we have 4 items and the knapsack bag has capacity of 15. Profit values are {10,12,20,18} and corresponding weights are {2,4,5,9}. Draw annotated state space tree with cost and bound values.

[Undated]

*Tip: Key formula: $\text{upper_bound} = \text{current_profit} + \text{remaining_capacity} * (\text{next item p/w ratio})$.*

Q2

10M

Given $n=4$ jobs with penalties $(p_1, p_2, p_3, p_4) = (5, 10, 6, 4)$, deadlines $(d_1, d_2, d_3, d_4) = (1, 2, 2, 1)$, and execution times $(t_1, t_2, t_3, t_4) = (1, 2, 1, 2)$. Find the actual cost and feasible jobs done using branch and bound technique.

[Apr 2023]

REPEATED TOPIC

Q3

10M

Apply the FIFO Branch and Bound approach to solve the 0-1 Knapsack Problem. Packages: weight=[2,3,4,5], value=[40,50,70,80]. Contrast Backtracking and Branch & Bound paradigms in terms of search strategy, pruning mechanism, efficiency, and applicability.

[Apr 2025]

Module 7: Graph Algorithms — Shortest Paths & Network Flow

CO3 | 12 Questions

Floyd-Warshall All-Pairs Shortest Path | Bellman-Ford Single Source | Ford-Fulkerson Max Flow | Edmonds-Karp Algorithm | Push-Relabel Algorithm

REPEATED TOPIC

Q1

10M

Find all pairs shortest path for the following weighted digraph using Floyd-Warshall's Algorithm. Nodes: A, B, C, D, E with given weights. Analyse time complexity. Illustrate the tracing of the algorithm.

[Undated]

Tip: Floyd-Warshall appeared 6/7 papers! Know $D^k[i][j] = \min(D^{k-1}[i][j], D^{k-1}[i][k] + D^{k-1}[k][j])$.

REPEATED TOPIC

Q2

10M

Perform step-by-step process of finding shortest path distance between every pair of vertices using Floyd-Warshall algorithm from the given directed weighted graph (vertices 1,2,3,4 with given edge weights).

[May 2024 A]

REPEATED TOPIC

Q3

10M

Find all-pairs shortest path for the given weighted digraph using Floyd-Warshall's Algorithm. Edges: B-A:10, A-C:12, C-B:20, C-D:32, A-D:60, E-A:7. Analyse time complexity. Detect whether a negative cycle exists using diagonal entries.

[Apr 2026]

REPEATED TOPIC

Q4

10M

Elaborate the procedure to find the shortest path from source to all other vertices in the given graph using Bellman-Ford algorithm. Graph has vertices A,B,C,D,E. Discuss how dynamic programming strategy is applied. State how Bellman-Ford detects negative weight cycles.

[Apr 2023]

Tip: Bellman-Ford appeared in 4 papers. Runs $(V-1)$ iterations. If relaxation still possible in V -th iteration, negative cycle exists.

REPEATED TOPIC

Q5

10M

Given directed weighted graph $G(V,E)$ with potentially negative weight edges, check whether Bellman-Ford can find shortest path. Determine possibility of negative cycles. Graph vertices 0,1,2,3.

[May 2024 B]

REPEATED TOPIC

Q6

10M

Solve Bellman-Ford single source shortest path computations. Starting vertex '1', destination vertex '7'. Graph has 7 vertices with various weights including negatives. Analyse complexity.

[May 2024 A] [May 2024 B]

REPEATED TOPIC

Q7

10M

Apply Ford-Fulkerson's algorithm on the following network flow graph to find the maximum flow from start vertex 'A' to sink vertex 'G'. In each iteration, show the graph traversal, the flow value to be augmented, and the residual flow network after augmenting.

[Undated]

Tip: Ford-Fulkerson appeared in ALL 7 papers. This is the #1 most tested topic!

REPEATED TOPIC

Q8

10M

Apply Ford-Fulkerson algorithm. Justify the need for adding flow along reverse edges in the residual graph. Given 3 networks (a,b,c) with augmented paths (0-1-3), (0-1-2-3), (0-1-2-3) respectively. Compute maxflow and determine which scenario justifies reverse edges.

[Winter 2022-23]

REPEATED TOPIC

Q9

10M

Apply Ford-Fulkerson — max flow from source 's' to sink 't'. Network with nodes s, a, b, c, d, e, t and capacities a-d:10, a-c:3/13/25, b-c:9/8/18, etc. Show each augmenting path and residual graph.

[Apr 2023] [May 2024 B] [Apr 2025] [Apr 2026]

REPEATED TOPIC

Q10

10M

Write Edmonds-Karp's algorithm, compute the maximum flow in network N (nodes S,A,B,C,D,T with given capacities: S-A:15, S-B:12, A-C:10, B-C:5, B-D:6, etc.) and analyse its complexity.

[May 2024 A]

Tip: Edmonds-Karp is Ford-Fulkerson with BFS for augmenting path. $O(VE^2)$.

Q11**10M**

Apply Push-Relabel algorithm to find the maxflow for the given graph. Source=0, sink=4. Graph edges: 0-1:12, 0-2:15, 1-2:7, 1-3:8, 2-4:6, 3-4:5.

[Winter 2022-23]

REPEATED TOPIC**Q12****10M**

A logistics company wants to maximize shipment flow using Ford-Fulkerson from source s to sink t through cities u, v, w, x, y . In first step, choose route with longest augmenting path. In second step, choose route with highest residual capacity. Show original graph, augmenting paths, flow network, and residual graphs after each step.

[Apr 2025]

Module 8: String Matching Algorithms

CO3 | 9 Questions

KMP Algorithm (LPS / Failure Function) | Rabin-Karp Algorithm (Rolling Hash) | Naive String Matching

REPEATED TOPIC

Q1

10M

Apply an algorithm that constructs Longest Prefix Suffix (LPS) table for the pattern to find where it is plagiarized in a given text. List the algorithm and trace steps for pattern 'ABABAD' and text 'ABCABABABAD'. Also return the position where the pattern is found. Analyze time complexity.

[Undated]

Tip: KMP appeared in 6/7 papers. Always build LPS table first!

REPEATED TOPIC

Q2

10M

Construct the Longest Prefix Suffix (LPS) table for pattern '121231213'. What is the length of the longest proper prefix and suffix? If characters from indexes 0 to 6 match but 7th does not, which character should next comparison start from? Compare time complexity with Naive pattern-matching.

[Winter 2022-23]

REPEATED TOPIC

Q3

5M

Interpret steps of KMP_next array and the KMP algorithm. Compute the KMP_next array of Text=ABCCDDAEFG and pattern=CDD.

[May 2024 A]

REPEATED TOPIC

Q4

10M

Assume you have text T and pattern P. Write an algorithm to report the index of the first occurrence (if any) of P in T from the pi-table of PT where PT is the concatenation of P and T. If P does not occur in T, then print -1. Trace with a positive example.

[May 2024 B]

REPEATED TOPIC

Q5

10M

Apply KMP and Rabin-Karp algorithm for finding all occurrences of pattern P='ACGTACGT' within target DNA sequence T='GCACGTACGTGACGTACGTA'. For Rabin-Karp, consider A=1, C=2, G=3, T=4, base=4, mod=101. Compare efficiency by calculating total character comparisons.

[Apr 2025]

Q6

5M

Classify between KMP and Rabin-Karp algorithm — strength and weakness of complexity.

[May 2024 A]

REPEATED TOPIC

Q7

10M

For string matching with hash function $\text{Pattern}(P) \bmod q$ where $q=13$, how many spurious hits does the Rabin-Karp matcher encounter in Text $T=42637135421$ and $P=54$? Find number of shifts to find the pattern. Write pseudocode and discuss time complexity.

[Apr 2023]

Tip: Rabin-Karp appeared in 3 papers. Know rolling hash: $H_{\text{new}} = (d \cdot (H_{\text{old}} - T[s] \cdot h) + T[s+m]) \bmod q$.

REPEATED TOPIC

Q8

10M

Write pseudocode for Rabin-Karp algorithm. Trace for text $T=\text{adbadbacfe}g\text{ba}$ and pattern $P=\text{ba}$. Hash code: $a=1, b=2, \dots, d=8, q=13$. Demonstrate rolling hashing function.

[May 2024 B]

REPEATED TOPIC

Q9

10M

Rabin-Karp — Text $T='14159265358979323846'$ (first 20 digits of π), Pattern $P='59265'$. Base $d=10$, Prime $q=13$. Compute initial hash values, show rolling hash for next 3 shifts, identify spurious hits. Analyse average and worst-case complexity.

[Apr 2026]

Module 9: Geometric Algorithms

CO3 | 7 Questions

Convex Hull — Graham's Scan $O(n \log n)$ | Convex Hull — Jarvis March $O(nh)$ | Line Segment Intersection

REPEATED TOPIC

Q1

10M

Solve the following instance of Convex Hull using Graham's Scan algorithm $O(n \log n)$ for points $Q=\{P1(10,3), P2(6,1), P3(8,4), P4(2,2), P5(3,5), P6(1,6), P7(7,3), P8(6,7)\}$. Elaborate the step-by-step process of constructing the polygon.

[Undated] [Apr 2026]

Tip: Graham's Scan appeared in 5/7 papers! Same point set repeated. Sort by polar angle from lowest point.

REPEATED TOPIC

Q2

10M

Relate Graham's Scan and Jarvis March Algorithm steps and how to choose the best practices for computational Convex Hull. Illustrate with an example.

[May 2024 A]

REPEATED TOPIC

Q3

5M

Apply Graham's Scan algorithm for point set $P=\{(1,2),(3,5),(4,4),(6,2),(7,6),(8,3),(2,7),(5,1)\}$. Illustrate step-by-step execution showing intermediate hull formations. Compare computational efficiency of Graham's Scan and Jarvis' March in terms of time complexity.

[Apr 2025]

REPEATED TOPIC

Q4

10M

Write the pseudocode for Jarvis-march algorithm and trace it for the given points $p1...p7$ provided in the diagram.

[May 2024 B]

Tip: Jarvis March: $O(nh)$ where h = number of hull points. Repeatedly find most counterclockwise point.

REPEATED TOPIC

Q5

10M

Apply Graham's Scan to find the convex hull of points $(2,1)(3,3)(5,4)(-2,2)(-3,1)(-4,3)(-1,-2)(-3,-4)(2,-3)(4,-4)$ in counter clockwise direction.

[Apr 2023]

Q6**10M**

Design an algorithm for checking whether the two given line segments intersect or not. Trace and find if line segment given by p_1q_1 and another line segment given by p_2q_2 intersect. Check 4 pairs: (i) $p_1(2,10)$ $q_1(7,2)$ $p_2(4,15)$ $q_2(13,3)$, (ii) $p_1(6,4)$ $q_1(10,16)$ $p_2(8,30)$ $q_2(13,6)$, etc.

[Undated]

Tip: Use cross product to determine orientation. 4 cases: general and collinear.

Q7**5M**

From the following set of line segments, find all the pairs that intersect, using a more efficient method than the Brute Force approach. Segments: $\{(2,8),(6,9)\}$, $\{(5,7),(10,7)\}$, $\{(3,5),(8,4)\}$, $\{(7,6),(12,12)\}$, $\{(9,10),(16,3)\}$, $\{(14,7),(17,9)\}$.

[Winter 2022-23]

Module 10: Randomized Algorithms & NP-Completeness

CO4/CO5 | 9 Questions

Randomized Quicksort | Hiring Problem | NP, NP-Hard, NP-Complete Classes | 3-CNF SAT | Clique Decision Problem | Vertex Cover

REPEATED TOPIC

Q1

10M

Apply Randomized Quicksort algorithm by selecting the random pivot to trace the steps to sort in detail. Discuss how the average time complexity of randomized quicksort is better than the worst-case time complexity of deterministic quicksort. Elements: [99,34,46,22,60,23,15,14,33,20].

[Undated] [Apr 2026]

Tip: Same array [99,34,46,22,60,23,15,14,33,20] appeared in 3 papers!

REPEATED TOPIC

Q2

10M

Narrate how Randomized Quick Sort algorithm differs from Normal Quick Sort. Write an algorithm and solve the randomized quick sort algorithm with an example.

[May 2024 A]

REPEATED TOPIC

Q3

10M

Apply Randomized Quicksort algorithm on [9,4,7,2,6,3,8]. Show step-by-step execution including selection of random pivots, partitioning, and recursive calls. Analyse expected time complexity and discuss how randomization helps avoid worst-case.

[Apr 2025]

Q4

10M

Design a randomized algorithm for the hiring problem. Consider cost to interview as c_i per candidate and cost to hire as c_h . Analyse performance and prove that when n candidates are presented in random order, algorithm Hire-Assistant has a hiring cost of $O(c_h \ln n)$.

[Apr 2023] [Winter 2022-23]

REPEATED TOPIC

Q5

10M

Provide a summarized explanation of the complexity classes P, NP, NP-Hard, and NP-Complete, highlighting their role in assessing the hardness of real-world computational problems. Additionally, demonstrate the NP-completeness of the 3-SAT problem by formally reducing SAT to 3-SAT.

[Apr 2025] [May 2024 A]

Tip: NP-completeness appeared in 5 papers. Know the standard reductions: SAT $\rightarrow$ 3-SAT $\rightarrow$ Clique.

REPEATED TOPIC

Q6

10M

Prove that 3-CNF (Conjunctive Normal Form) SAT is NP complete. Use the satisfiability formula given below to prove NP completeness. Illustrate the three steps for conversion of ϕ to ϕ'' . For $\phi = (x_4 \text{ AND } ((x_1 \rightarrow x_2) \text{ AND } (x_3 \leftrightarrow x_1)))$.

[Undated]

REPEATED TOPIC

Q7

10M

Prove clique decision problem is NP-complete with the help of the following 3-CNF Sat: $(a \vee \bar{b} \vee c) \text{ AND } (\bar{a} \vee b \vee c) \text{ AND } (a \vee b \vee c)$.

[Apr 2023] [Winter 2022-23]

REPEATED TOPIC

Q8

10M

Prove that the Clique Decision Problem is NP-Complete using 3-CNF SAT: $(a \vee \bar{b} \vee c) \text{ AND } (\bar{a} \vee b \vee c) \text{ AND } (a \vee b \vee c)$. Provide summarized explanation of Vertex Cover Problem, its NP-Complete proof, and 2-approximation algorithm.

[Apr 2025] [Apr 2026]

Q9

10M

Brief the various complexity classes and illustrate the conversion of SAT problem to 3-CNF-SAT problem.

[May 2024 A]

TOP 10 PREDICTED QUESTIONS — EXAM 2026

Based on April 2026 format: Answer ALL 10 Questions × 10 Marks = 100 Marks. These predictions are based on frequency analysis of 7 past papers.

#1

Ford-Fulkerson / Max Flow

Appeared in EVERY single paper. Guaranteed to come.

7/7

Apply Ford-Fulkerson's algorithm on the given network flow graph to find the maximum flow from start vertex 's' to sink vertex 't'. In each iteration, show the graph traversal performed to find the augmenting path, the flow value to be augmented, and the residual flow network obtained after augmenting the flow.

#2

Knapsack (Fractional / 0-1)

Appeared in EVERY paper — either Greedy (Fractional) or DP/Branch&Bound; (0-1).

7/7

Design a Fractional Knapsack algorithm using Greedy approach. Given profits and weights, show profit/weight ratio calculations and item selection order to find maximum profit for capacity W. Why is Fractional Knapsack solved using Greedy while 0/1 Knapsack cannot be?

#3

Floyd-Warshall (All-Pairs Shortest Path)

One of the most consistently tested topics. Know the D-matrix iterations.

6/7

Find all-pairs shortest path for the given weighted digraph using Floyd-Warshall's Algorithm. Perform D^0 through D^n matrix iterations step by step. Analyse time complexity $O(n^3)$. Detect whether a negative cycle exists using diagonal entries.

#4

KMP String Matching + LPS Table

LPS table construction and KMP trace are always tested together.

6/7

Apply KMP algorithm — construct the Longest Prefix Suffix (LPS / failure function) table for given pattern. Trace pattern matching on given text. State all positions where pattern is found. Compare time complexity $O(n+m)$ with Naive $O(nm)$ algorithm.

#5

Randomized Quicksort

Same element array [99,34,46,22,60,23,15,14,33,20] appeared in 3 papers!

5/7

Apply Randomized Quicksort algorithm by selecting random pivot to trace the steps to sort [99,34,46,22,60,23,15,14,33,20] in detail. Discuss how average time complexity $O(n \log n)$ of randomized quicksort is better than worst-case $O(n^2)$ of deterministic quicksort.

#6

Convex Hull (Graham's Scan)*Graham's Scan with same point set P1(10,3)...P8(6,7) appeared in 2+ papers.*

5/7

Consider a set of points P in a XY plane. Construct a smallest polygon enclosing all the points. Design an $O(n \log n)$ time algorithm and trace it using points P1(10,3), P2(6,1), P3(8,4), P4(2,2), P5(3,5), P6(1,6), P7(7,3), P8(6,7). Elaborate the step-by-step process.

#7

NP-Completeness / 3-CNF SAT / Clique*Either prove NP-completeness or explain P/NP/NP-Hard/NP-Complete classes.*

5/7

Provide a summarized explanation of the complexity classes P, NP, NP-Hard, and NP-Complete. Additionally demonstrate the NP-completeness of 3-SAT / Clique Decision Problem by formally performing the appropriate reduction. Discuss the significance of NP-complete problems.

#8

Recurrence Relations + Master Theorem *$T(n)=3T(n/4)+n^2$ appeared in 2 papers. Master Theorem Case 3.*

4/7

Obtain an upper bound for $T(n) = 3T(n/4) + n^2$ using the recursive tree method. Draw the recursion tree, compute the cost at each level, and verify the result using Master's theorem.

#9

Divide & Conquer Max Subarray*Appeared in Apr 2026 format. Know how to compute left/right/cross subarray sums.*

4/7

Design the pseudocode for Divide and Conquer Maximum Subarray. Apply the algorithm to find the contiguous subarray with the largest sum in $A = [-2, 5, -1, 3, -4, 7, -2, 1]$. Show the left sum, right sum, and cross sum for each recursive division. State starting and ending indices and the sum.

#10

TSP — Held-Karp DP Algorithm*Growing trend — DP-based TSP is becoming the standard format.*

4/7

Solve the Travelling Salesman Problem for given 4-city distance matrix using Held-Karp (Dynamic Programming) algorithm. Show all $g(S,i)$ calculations step by step, determine the minimum cost tour, and trace back the optimal path. Compare time complexity $O(2^n * n^2)$ with brute force $O(n!)$.

QUICK REFERENCE — ALGORITHMS CHEAT SHEET

Algorithm	Complexity	Key Formula / Idea
Floyd-Warshall	$O(n^3)$	$D^k[i][j] = \min(D^{k-1}[i][j], D^{k-1}[i][k] + D^{k-1}[k][j])$
Bellman-Ford	$O(VE)$	Relax all edges $(V-1)$ times. V -th relaxation = negative cycle.
Ford-Fulkerson	$O(E * \text{max_flow})$	BFS/DFS to find augmenting path, update residual graph, repeat.
Edmonds-Karp	$O(VE^2)$	Ford-Fulkerson with BFS only. Guaranteed polynomial.
KMP	$O(n+m)$	Build LPS failure function in $O(m)$, then scan text in $O(n)$.
Rabin-Karp	$O(n+m)$ avg	Rolling hash: $H_{\text{new}} = (d*(H_{\text{old}} - T[s]*h) + T[s+m]) \bmod q$.
Graham's Scan	$O(n \log n)$	Sort by polar angle. Use stack, pop if right turn (CW).
Jarvis March	$O(nh)$	Repeat: find most CCW point from current. Stop when back at start.
Matrix Chain DP	$O(n^3)$	$m[i][j] = \min \text{ over } k \text{ of } m[i][k] + m[k+1][j] + p[i-1]*p[k]*p[j]$.
Held-Karp TSP	$O(2^n * n^2)$	$g(S,i) = \min_{k \in S - \{i\}} [g(S - \{i\}, k) + \text{cost}(k,i)]$.
Fractional Knapsack	$O(n \log n)$	Sort by profit/weight ratio. Take greedily. Fraction of last item OK.
Karatsuba	$O(n^{1.585})$	3 multiplications instead of 4: $ac, bd, (a+b)(c+d) - ac - bd$.
Randomized Quicksort	$O(n \log n)$ avg	Pick random pivot. Expected $T(n) = O(n \log n)$ regardless of input.
Master Theorem	3 Cases	Case 1: $f(n) = O(n^{\log_b(a)-\epsilon}) \rightarrow T = \Theta(n^{\log_b(a)})$. Case 2: $f(n) = \Theta(n^{\log_b(a)}) \rightarrow T = \Theta(n^{\log_b(a)} \log n)$. Case 3: $f(n) = \Omega(n^{\log_b(a)+\epsilon}) \rightarrow T = \Theta(f(n))$.

All the best for your exam! — BCSE204L DAA PYQ Bank